MIL-D-11637A (Ord)
1 March 1954
SUPERSEDING
MIL-D-11637 (Ord)
13 December 1951

MILITARY SPECIFICATION

DRIVES, BALLISTIC, M4 (T23), M3 (T23E1) AND T23E2

1. SCOPE

1.1 This specification covers three types of instruments known as Drive, Ballistic, M4; Drive, Ballistic, M3; and Drive, Ballistic, T23E2. These ballistic drives, in the order mentioned above, are used to introduce superelevation into the light, medium and heavy tanks respectively.

2. APPLICABLE SPECIFICATIONS, STANDARDS, DRAWINGS, AND PUBLICATIONS

2.1 The following specification and drawings, and specifications and drawings listed thereon, of the issue in effect on the date of invitation for bids, form a part of this specification:

SPECIFICATIONS

Military

MIL-P-13008 - Packaging of Drive, Ballistic, T23E2

MIL-P-12412 - Packaging of Drive, Ballistic, T23 & T23E1

U. S. Army

51-7 - Instruments, Fire-Control; General Specification Governing the Manufacture and Inspection of

DRAWINGS

Ordnance Corps

F7649132 - Drive, Ballistic, (T23) M4

F7649131 - Drive, Ballistic, (T23E1) M3

F7649133 - Drive, Ballistic, T23E2

(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

FSC 1220

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3. REQUIREMENTS

3.1 General requirements common to all fire control instruments as contained in Specification 51-7 shall be mandatory in so far as they are applicable.

3.2 Materials shall be in accordance with the applicable drawings and specifications forming part of this specification.

3.3 Accuracy.

3.3.1 The distance between the two extreme bore centers of the arms, and support, as well as between the housing shaft bore center and the split bore center in the base of the ballistic unit assemblies, shall comply with the following drawing requirements:

BALLISTIC DRIVE	ARM	SUPPORT	BALLISTIC UNIT ASSEMBLY
(T23) M4	C7654917		F7652238
(T23E1) M3	C7649219		F7649135
T23E2		F7688566	F7649227

3.3.2 With the coupling located to the left of the ballistic unit in a fixed position, any two ranges selected at random from each range scale shall produce an angular change at Coupling B7649156 (see 6.1) corresponding to the elevation values within plus or minus 0.2 mil as tabulated on the range scale drawings.

3.4 Alignment.

3.4.1 With the vial in the level position, the graduations on all range scales shall indicate zero.

3.4.2 With the vial in the level position, alignment of couplings shall be in accordance with all notes outlined on the following drawings:

BALLISTIC DRIVE	BALLISTIC UNIT ASSEMBLY
(T23) M4	F7652238
(T23E1) M3	F7649135
T23E2	F7649227

3.5 Backlash.- With the stud of the Connector Assemblies C7652212 and C7633019 for Ballistic Drives (T23) M4 and (T23E1) M3 respectively, and Coupling B7649018 for Ballistic Drive T23E2, held rigidly, the backlash in each instrument shall not exceed 0.1 mil as measured on Coupling B7649156, (see 6.1).

3b6 Temperature.- The temperature requirement and test shall not apply.

3.7 Scale illumination.- With the scale illuminated by a 24 volt direct current source, and then by the emergency Instrument Light M30 or equivalent, the scale graduations in both instances shall appear clearly defined when observed in a dark room.

3.8 Workmanship shall be in accordance with Specification 51-7.

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4. SAMPLING AND INSPECTION

4.1 Unless otherwise specified, each ballistic drive shall be inspected for conformance with this specification as specified in Specification 51-7.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking.- Packaging, packing and marking shall be in accordance with Specification MIL-P-13008 or MIL-P-12412, as applicable.

6. NOTES

6.1 Informational data.- Backlash in Coupling Assembly B7649155 (that is, between Coupling B7649156 and Plate B7652107) will be absorbed by the spring in the head of Periscope (T35) M20.

6.2 Ordering data.- Invitation for bids and contracts should specify the title, number and date of the specification, and type of packing.

NOTICE: This specification, together with specifications and drawings pertaining to it and bearing a "NOTICE" of similar restrictions, is intended for use only in connection with procurement by the United States Government; and shall not be reproduced either wholly or in part except when authorized in connection with Government procurement, nor be used for any other purpose except when specifically authorized by the Chief of Ordnance.

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Army - Ordnance Corps

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

S/N-0102-014-1801 C-25254

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